HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F70LB

LIFE INSURANCE MATHEMATICS B

IN-PERSON EXAMINATION

April Examination and Assessment Diet 2024

Duration: 2 hours

Total marks available: 60

SOLUTIONS

Question 1

You are given the following information in respect of two lives (x) and (y):

$$\ddot{a}_{xy} = 8$$
$$\ddot{a}_{x} = 9$$
$$\ddot{a}_{y} = 10$$
$$Force\ of\ interest\ \delta = 5\%\ \text{each year}$$
$$\mu_{x+t} = 0.01$$
$$\mu_{y+t} = 0.03$$

Calculate the following:

(a) $\bar{a}_{x|y}$

$$\bar{a}_{x|y} = \bar{a}_{y} - \bar{a}_{xy} = (\ddot{a}_{y} - 0.5) - (\ddot{a}_{xy} - 0.5) = 10 - 8 = 2$$

[1 mark for $\bar{a}_{x|y} = \bar{a}_{y} - \bar{a}_{xy}$; 1 mark correct answer]

(b) $\bar{A}_{\overline{xy}}$

$$\bar{A}_{\overline{xy}} = 1 - \delta\bar{a}_{\overline{xy}}$$ [0.5 marks]
$$\bar{a}_{\overline{xy}} = \bar{a}_{x} + \bar{a}_{y} - \bar{a}_{xy}$$ [0.5 marks]
$$\bar{A}_{\overline{xy}} = 1 - 0.05\,((9 - 0.5) + (10 - 0.5) - (8 - 0.5)) = 0.475$$ [1 mark]

(c) $\mu_{x+t\,:\,y+t}$=0.01+0.03=0.04 [1 mark]

Question 2

(a) Calculate the crude death rate for Country A in 2016 and Country B in 2021.

Country A in 2016 CDR = 219/0.73 = 300 [1 mark]
Country B in 2021 CDR = 1414/5.47 = 258.5 [1 mark]

(b) Calculate the standardised death rate for Country A in 2016.

(250*10+305*10+344*20+300*20)/60 = 307.2

(1 mark for correct formula; 1 further mark for numerically correct answer.)

(c) Calculate the standardised mortality ratio for Country B in 2021.

Expected deaths = 300 * 5.47 = 1641 (or 300*1.23+300*1.3+300*1.38+300*1.56)
SMR = 1414/1641 = 86%

(1 mark for correct formula; 1 further mark for numerically correct answer.)

(d) State with reasons two types of selection which are relevant to the comparison.

Class selection is differences in mortality based on some permanent feature of the population. It is relevant here because were are looking at two different countries. [1 mark]

Time selection is differences in mortality rates linked to lives being in different time periods. We are looking at two different years here. [1 mark]

[Total: 8 marks]

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Question 3

One mark for each of the highlighted cells. Credit given where errors carried forward.

1. Unit Fund

Year	Premium	Alloc Factor	B\O spread	Premium Alloc'd	Policy Fee	Fund return	Year-end Fund (1)	FMC rate	FMC	Fund after FMC	MC	Year-end Fund (2)
0												0.00
1	£2,000.00	0.97	4%	1862.40	25.00	1%	1855.77	1.5%	27.84	1827.94	11.72	1816.22
2	£2,000.00	1.02	4%	1958.40	25.00	-3%	3637.13	1.5%	54.56	3582.57	0.00	3582.57

2. Sterling Fund

Year	Unalloc Premium	Expense	Policy Fee	Interest	FMC	Fund mangement cost	MC	q	Death Cost	SFC_t
0										
1	137.60	250.00	25.00	-0.87	27.84	9.08	11.72	0.002	2.34	-60.1
2	41.60	20.00	25.00	0.47	54.56	17.91	0.00	0.003	0.00	83.7

3. Profit Test

Year	SFC_t	{t-1}_p_x	sigma_t
1	-60.14	1.000000	-60.14
2	83.71	0.998000	83.54

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Question 4

2 marks given for any sensible, correct answer. One mark given if the answer shows partial understanding.

Sample answer sufficient for 2 marks: "In a DC scheme the member has some certainty over contributions, but the risks of investment returns before retirement, annuity prices at retirement, fees and (if an annuity is not purchased) longevity and investment risk after retirement all fall on the member. In a DB scheme, the member has certainty over how their pension will be calculated. They may still have some risks (eg, inflation, failure of the pension scheme) but these are normally much less than DC. The DB sponsor carries most of the risk."

Question 5

A life office uses a competing risk model with three states labelled 0, 1 and 2. The transition intensities are $\mu_t^{01} = 1$ and $\mu_t^{02} = 2$. All other transition intensities are zero.

(a) State the three assumptions normally used for a Markov multi-state model.

(i) All transition intensities depend only on the the current time t, not any other aspect of the life's history. [1 mark]

(ii) For any small time period dt, we have ${}_{dt}p_{x+t}^{ij} = \mu_{x+t}^{ij} dt + o(dt)$ [1 mark]

(iii) The probability of making 2 transitions in a short time period dt is $o(dt)$ [1 mark]

(b)

(i) Prove from first principles that in this model

$$\frac{d}{dt}\left({}_tp_x^{00}\right) = -3\ {}_tp_x^{00}$$

${}_{t+dt}p_x^{00} = {}_{dt}p_{x+t}^{00}\ {}_tp_x^{00}$ [1 mark]

$= (1 - {}_{dt}p_{x+t}^{01} - {}_{dt}p_{x+t}^{02})\ {}_tp_x^{00}$ [0.5 marks]

$= (1 - \mu_{x+t}^{01} dt - \mu_{x+t}^{02} dt + o(dt))\ {}_tp_x^{00}$ using assumption (ii) [0.5 marks]

$= {}_tp_x^{00} - 3\ {}_tp_x^{00} dt + o(dt)$ using $\mu_t^{01} = 1$ and $\mu_t^{02} = 2$ [0.5 marks]

$= {}_tp_x^{00} - 3\ {}_tp_x^{00} dt + o(dt)$

Rearranging: $({}_{t+dt}p_x^{00} - {}_tp_x^{00})/dt = -3\ {}_tp_x^{00} + o(dt)/dt$ [1 mark]

Taking the limit as $dt \to 0$ gives the required result since the LHS is the definition of the derivative of ${}_tp_x^{00}$
[1 mark]

(ii) State, with reasons, which assumptions from part (a) were needed in your proof in part (b)(i) and which were not needed.

We used assumption (ii) in the third line above. [0.5 marks]
There is no way to make 2 transitions in this model, so we did not need assumption (iii). [0.5 marks]
For the same reason, we did not really need assumption (i). If the life is state 0 then we know its entire history. (It has always been in state 0.)
[0.5 marks]

(c) State the Kolmogorov forward equations for ${}_tp_x^{01}$ and ${}_tp_x^{02}$.

$$\frac{d}{dt}\left({}_tp_x^{01}\right) = {}_tp_x^{00}$$
$$\frac{d}{dt}\left({}_tp_x^{02}\right) = 2\,{}_tp_x^{00}$$

[0.5 marks each]

(d) Use your answers to (b) and (c) in an Euler scheme with step size $h = 0.1$ to estimate ${}_{0.2}p_x^{00}$.

${}_0p_x^{00} = 1;\ {}_0p_x^{01} = 0$ [1 mark] $({}_0p_x^{02} = 0\ is\ not\ needed)$

So $\frac{d}{dt}\left({}_tp_x^{00}\right)_{t=0} = -3$

And ${}_{0.1}p_x^{00} \approx 1 + 0.1 \times \frac{d}{dt}\left({}_tp_x^{00}\right)_{t=0} = 0.7$ [0.5 marks]

$\frac{d}{dt}\left({}_tp_x^{01}\right)_{t=0} = {}_0p_x^{00} = 1$

And ${}_{0.1}p_x^{01} \approx 0 + 0.1 \times \frac{d}{dt}\left({}_tp_x^{01}\right)_{t=0} = 0.1$ [0.5 marks]

$\frac{d}{dt}\left({}_tp_x^{01}\right)_{t=0.1} = {}_{0.1}p_x^{00} = 0.7$

Hence ${}_{0.2}p_x^{01} \approx 0.1 + 0.1 \times \frac{d}{dt}\left({}_tp_x^{01}\right)_{t=0.1} = 0.1 + 0.1 * 0.7 = 0.17$ [1 mark]

(e) Use your answer to (c) to solve ${}_tp_x^{01}$ and hence calculate the exact value of ${}_{0.2}p_x^{01}$.

Solving $\frac{d}{dt}\left({}_{t}p_x^{00}\right) = -3\,{}_{t}p_x^{00}$ gives ${}_{t}p_x^{00} = e^{-3t}$ [1 mark]

Combining with (c), $\frac{d}{dt}\left({}_{t}p_x^{01}\right) = {}_{t}p_x^{00} = e^{-3t}$ [0.5 marks]

Integrating gives ${}_{t}p_x^{01} = [(-1/3)e^{-3s}]_0^t = 1/3 - (1/3)e^{-3s}$ [1 mark]

Hence ${}_{0.2}p_x^{01} = 0.150$ [0.5 marks]

(f) Briefly explain why we usually use numerical methods (not exact solutions) to find probabilities in a multi-state model.

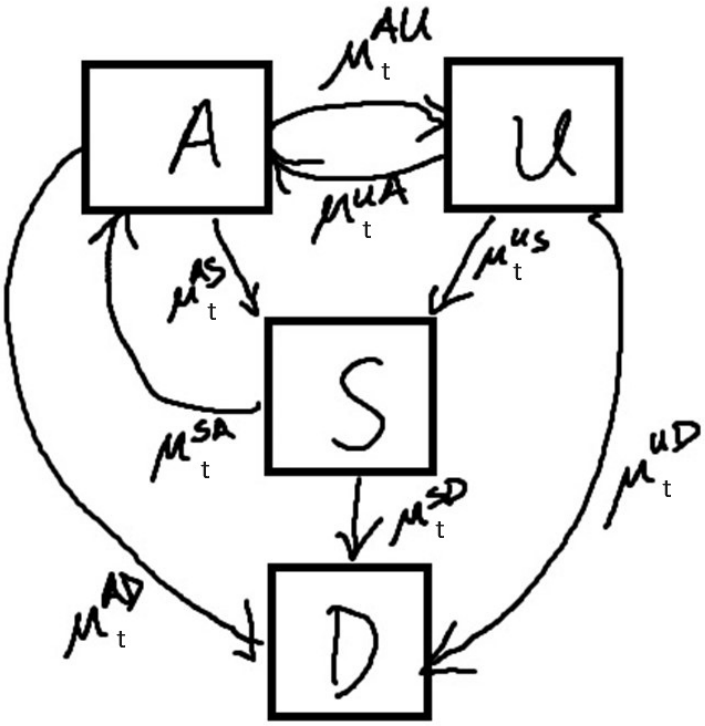
1 mark for any sensible answer such as:

- We will usually work with estimated transition intensities that don't follow a formula and cannot be solved analytically

[Total 13 marks]

Question 6

(a)



3 marks – All correct

2 marks – One minor error, eg missing transition labels or one missing/extra arrow

1 mark – Diagram shows all four states and most arrows (5 out of 8) are correct

0 marks – One of the states is missing or the arrows are mostly incorrect

(b) One half mark for each bullet point except where stated, up to a maximum of 4 marks:

- The transition intensities could be improved.
- It is only worth improving them if it would make a material difference to the answers (model outputs).
- Transition A to U:
    - It makes sense that it reduces over time because the longer someone has been in stable employment, the less likely they are to lose their job.
    - It doesn't take account of reaching retirement age.
    - It is strange that it falls to zero after 10 years.
- Transition A to S and U to S:
    - These are the same (0.03), but perhaps if someone does not have a job they are more likely to get sick.
    - They are constant, but the risk of sickness might increase over time.
- Transition S to U is zero:

  - This might make sense in terms of the policy if the benefit only stops if they policyholder returns to work.
  - But it is not necessarily realistic in terms of the underlying process. Someone could get well, but not immediately find a job to return to.
- Transitions to the dead state:
  - It makes sense that they increase with time (ie, with age)
  - They increase too rapidly with age
  - They only make sense if all policyholders are the same age, otherwise you will have very different rates for people the same age.
- Transition from S to A:
  - This is constant, but would probably decrease the longer the life had been in the sick state.
- If any benefits continue beyond the initial 10-year policy term then some of the transition intensities make no sense.
- Improvements:
  - Use a semi-Markov model, ie transition intensities that depend on time in a state (especially state S, perhaps also state A or U) [1 mark]
  - Transition intensities into state S and D should depend on age, not time since the policy was taken out
  - Transition intensities into the dead state (or from A to U) could be improved by having them increase (decrease) at a more realistic rate.

(c)

(i) The EPV of a premium of £1000 paid whenever the life is in the Active state [1 mark]

(ii) The EPV of a £100k lump sum [1 mark]
…paid immediately on death from any state [1 mark]

(iii) The EPV of a waiver of premium benefit [0.5 marks]
… with a waiting period of 1 year [0.5 marks]
…paid for the first period of of unemployment only [0.5 marks]
…on condition that the policyholder has never been sick [0.5 marks]

(iv) The EPV of an income of £25k/year paid to the policyholder while sick [0.5 marks]
…for any period of sickness that starts during the policy term [0.5 marks]
…with a deferred period of 0.25 years, ie 3 months [0.5 marks]
… and paid for life (or until recovery) [0.5 marks]

Commentary sheet for F70LB/F71BF Exam

Module Code & Title: F70LB/F71BF Life Insurance Maths B / 2
Semester: 1
Exam/CA ratio: 70:30
IFoA Subject: 25% of CM1

Question 1: Calculating joint life annuities/assurances, transition intensities
CM1 Syllabus Objective 5 (various)
Standard style of question, but unseen. Should be straightforward for most students. Have given them the underlying figures instead of requiring them to use orange books.

Question 2: Single figure indices and selection
Single figures indices are on our syllabus, not IFoA. Selection features outside CM1.
Unseen, but a simple calculation.

Question 3: Emerging cashflows
CM1 Syllabus Objective 6.3, 6.4, 6.5
Unseen, although similar questions with different parameters are often asked. Well-prepared students should be able to score well.

Question 4: Pensions
On our syllabus not IFoA. Only 2 marks.
Simple knowledge question.

Question 5: Multi state models, Euler schemes
CM1 Syllabus Objective 5 (various)
Unseen. Students will find this a little more challenging. They have seen similar proofs many times, but sometimes struggle with unseen special cases, like part (b). The Euler scheme should be straightforward. Even though part (e) is a simple calculation, students will find it unexpected.

Question 6: Multi state models, Euler schemes
CM1 Syllabus Objective 5 (various)
Unseen. Students will find parts (b) and (c) difficult.